

Oil/Gas Project

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Executive Summary

This project analyzes how oil and gas firms adjusted production during the COVID-19 shock in 2020, with a focus on shut-in decisions for horizontal wells. The analysis highlights clear variation in shut-in behavior across wells and firms, suggesting that production levels and operational characteristics play an important role in these decisions. The results indicate that firms did not respond uniformly to the market downturn, and that well-specific factors are key to understanding both the likelihood and duration of shut-ins. These findings provide useful insights into how energy producers manage operations under extreme economic conditions and can inform future modeling and decision-making in similar scenarios.

Project Description

This study is an observational analysis using monthly well-level production data from 2020. The project is in the analysis stage and focuses on examining the shut-in behavior of oil and gas wells in response to the COVID-19 shock, which led to a sharp decline in energy demand and prices. The analysis is restricted to horizontal wells to ensure comparability and centers on firms operating in the Bakken region, specifically Whiting and XTO.

The objective of the study is to identify which wells were shut in and to determine how factors such as production levels, wastewater output, and geographic location influence shut-in decisions. The analysis will first summarize shut-in patterns over time, then estimate probit models for shut-in decisions after February 2020, and finally apply duration models to examine how long wells remain shut in. The results are intended to provide insight into firm and well-level responses to economic shocks in the energy sector.

The economic context of this analysis is critical. In April 2020, Whiting Petroleum Corporation, a major operator in the Bakken region, filed for Chapter 11 bankruptcy following the collapse in oil prices during the COVID-19 pandemic. This highlights the severity of the financial pressures faced by firms in this period. As a result, shut-in decisions were likely influenced not only by production characteristics but also by financial constraints and broader firm-level considerations.

Research Questions

Question 1: Do shut-in behaviors differ between firms, particularly between Whiting and XTO in the Bakken region?

Question 2: How long do wells remain shut in before returning to production, and are shut-in durations influenced by firm behavior and well characteristics?

Statistical Questions:

Question 1: Can a probit model adequately explain shut-in decisions based on observable well characteristics?

Question 2: Is it statistically appropriate to combine Whiting and XTO wells into a single model, or should they be modeled separately?

Exploratory Data Analysis (EDA)

Data Overview

The data used in this analysis comes from a site that is maintained by the North Dakota State Government’s Department of Mineral Resources. They offer a monthly production report that lists production and operational details for oil and gas wells across North Dakota. We are specifically interested in the Bakken region. Data in these reports are provided by the companies running the wells via required reporting systems. Each record in the datasets represents a single well’s production for the month, which gives us a granular perspective relative to aggregated areas or overall company performance. We specifically pulled datasets from January of 2020 to December of 2021.

Table 1: Summary of Variables Used in the Analysis

Variable Name	Type	Description
api_wellno	Identifier	Unique well identifier
company	Categorical	Operator name
month	Time	Month of observation
date	Date	Date of observation
oil	Numeric	Monthly oil production (barrels)
gas	Numeric	Monthly gas production (MCF)
wtr	Numeric	Water production (wastewater)
days	Numeric	Number of production days
flared	Numeric	Gas flared
lat	Numeric	Latitude of well
long	Numeric	Longitude of well
well_name	Character	Well name (used to identify horizontal wells)
production	Numeric (derived)	Total production (oil + gas/6)
shut_in	Binary (derived)	Indicator for shut-in (<50% of previous month)
recovered	Binary (derived)	Indicator for recovery (>=75% of pre-shut production)

The report identifies these individual wells using a unique identifier (API well number). The company is listed, as well as geographic information, and production variables (oil, wastewater, and gas produced). These can be viewed in Table 1. To better serve our analysis, we created our own features. The first is production, which we defined as $\text{oil} + (\text{gas} / 6)$. This follows the industry standard of dividing gas, measured in MCF (thousand cubic feet), by 6 to convert the value into an energy output equivalent to a barrel of oil. We also created complementing features `shut_in` and `recovered`, which are described in Table 1.

A major limitation with this dataset is that there is no accompanying literature to describe the features. Important features such as oil, gas, and wtr (water produced) don’t have units. This is critical, as a difference in units can lead to wild misunderstanding if interpreted incorrectly. To move forward with this data, we had to make the assumption that the dataset is using the most common and conventional production units.

Visualizations

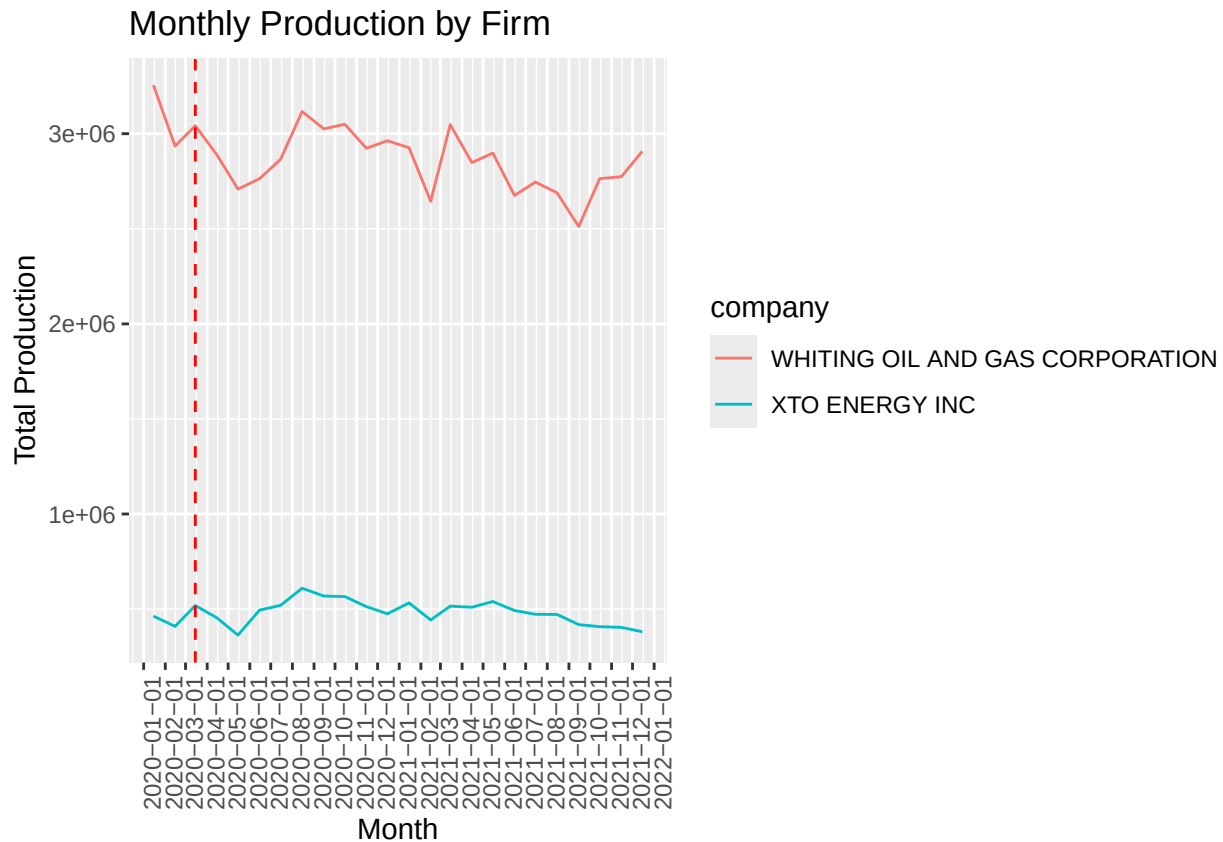


Figure 1: Monthly total production for Whiting and XTO over time, showing a sharp decline beginning in March 2020.

The observed separation in production levels suggests that Whiting operated a substantially larger or more productive portfolio of wells compared to XTO. As shown in Figure 1, this difference is important because firms with higher baseline production may face different economic incentives when deciding whether to shut in wells. In particular, larger producers may be more sensitive to price shocks due to higher operating costs and exposure to market fluctuations.

The temporary increases in production during 2020 may reflect partial operational recovery after the initial COVID-19 shock. While causality cannot be established, these patterns are consistent with improving demand conditions and stabilization in oil prices (Figure 1).

The sharp increase in shut-in rates immediately following March 2020 aligns with the collapse in oil demand and prices during the early stages of the COVID-19 pandemic. As shown in Figure 2, the similarity in shut-in patterns across firms suggests that both were responding to the same external market shock, although later divergence may reflect firm-specific strategies or financial constraints. This directly addresses the requirement to calculate monthly shut-in percentages for each firm.

As shown in Figure 3, most shut-in episodes lasted approximately one month, indicating that firms primarily used shut-in as a short-term response to unfavorable market conditions. This suggests that many wells remained economically viable but were temporarily taken offline due to low prices. However, a small number of wells experienced extended shut-in durations of up to 12 months, which may indicate wells with persistently low productivity or those affected by longer-term strategic or financial considerations.

Shut-in rates vary across production quartiles, with the highest rates observed among the lowest-producing

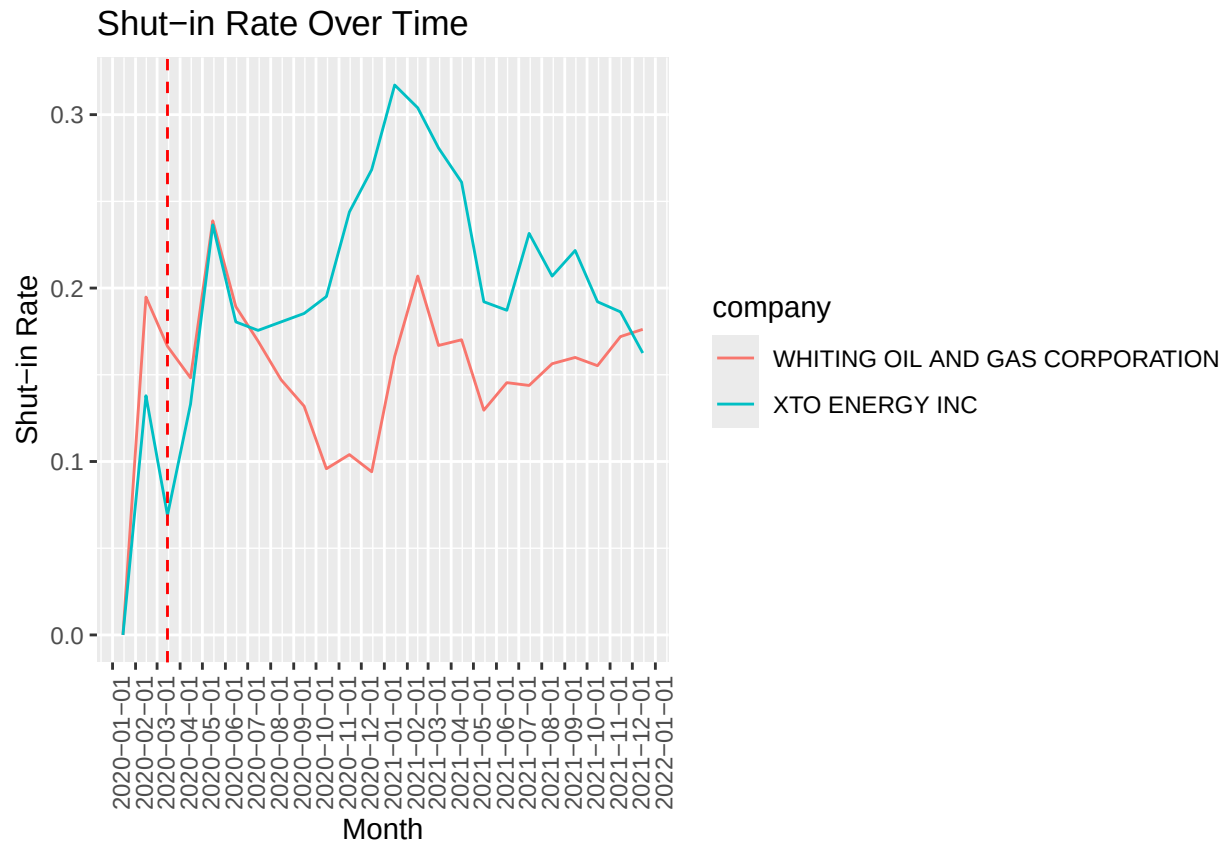


Figure 2: Monthly shut-in rate for Whiting and XTO, showing a sharp increase following the COVID-19 shock in early 2020.

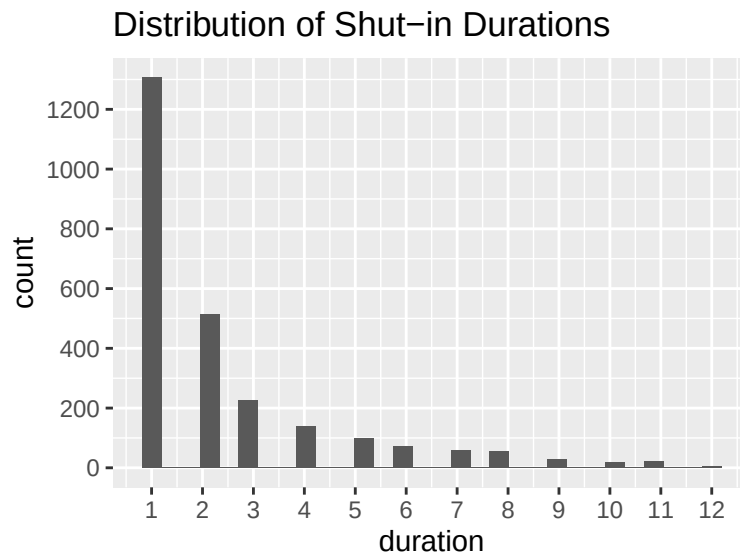


Figure 3: Distribution of shut-in durations across wells, showing that most shut-in episodes last approximately one month.

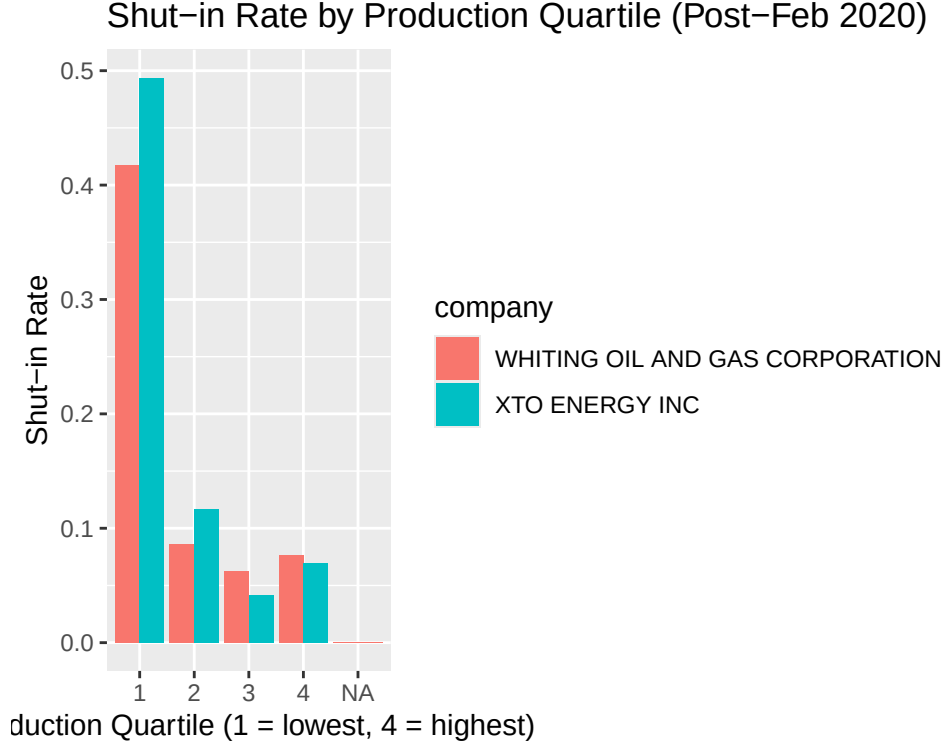


Figure 4: Shut-in rate by production quartile after February 2020, showing higher shut-in rates among lower-producing wells.

wells (Figure 4). This pattern suggests that firms prioritized keeping higher-output wells active while shutting in less productive wells during the COVID-19 downturn. While the relationship is not perfectly monotonic, the overall trend indicates that production level is an important determinant of shut-in decisions.

The log-scaled boxplot shows that production distributions overlap substantially across shut-in statuses (Figure 5), indicating that differences are not driven solely by extreme values. However, the higher frequency of shut-ins among lower-producing wells supports the hypothesis that firms selectively reduce output from less economically efficient wells during periods of economic stress.

Statistical Analysis

To formally evaluate the relationship between well characteristics and shut-in decisions, we created probit models, as recommended by the project brief. We also created logit models, which serve as robustness checks to assess whether the findings are sensitive to the choice of link function. To make a clear, evidence-based statement, we conducted hypothesis tests using the probit models. The hypotheses are as follows:

$$\begin{aligned}
 H_0 &: \text{Well characteristics have no effect on shut-in probability} \\
 H_1 &: \text{At least one characteristic affects shut-in probability}
 \end{aligned}$$

We fit separate models for each firm, as well as a combined model, to assess whether pooling the data is appropriate. It is also worth noting that a log transformation was applied to production and wastewater variables to reduce skewness and limit the influence of extreme values.

The probit model results in (Tables 2-4) indicate that shut-in decisions were driven by a combination of production factors and location, and across both firms the production levels and wastewater production

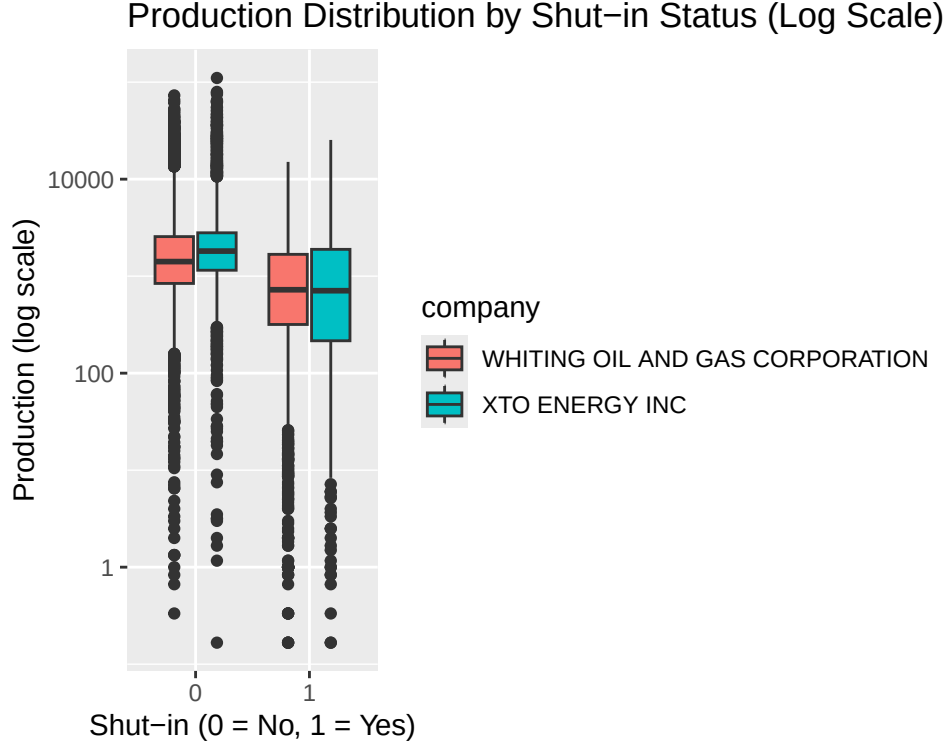


Figure 5: Production distribution by shut-in status (log scale), comparing production levels for shut-in and non-shut-in wells across firms.

Table 2: Whiting Probit Model Results

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-93.0884939	3.3289135	-27.963626	0.0000000	-99.5434199	-86.6548223
log_lag_production	0.5381687	0.0125902	42.744988	0.0000000	0.5144870	0.5623517
log_lag_wtr	-0.3667054	0.0069604	-52.684181	0.0000000	-0.3800215	-0.3535189
lat	0.3094862	0.0293843	10.532366	0.0000000	0.2522187	0.3672278
long	-0.7357687	0.0271866	-27.063623	0.0000000	-0.7883414	-0.6834071
month_factor2	0.1594331	0.0647218	2.463359	0.0137642	0.0316037	0.2875314
month_factor3	-0.3107098	0.0613771	-5.062310	0.0000004	-0.4314558	-0.1897177
month_factor4	-0.1968737	0.0601885	-3.270951	0.0010719	-0.3153203	-0.0780360
month_factor5	-0.0637708	0.0594850	-1.072049	0.2836982	-0.1807548	0.0537120
month_factor6	-0.2915996	0.0608537	-4.791810	0.0000017	-0.4114408	-0.1714457
month_factor7	-0.3147090	0.0611255	-5.148569	0.0000003	-0.4350242	-0.1941137
month_factor8	-0.3050611	0.0609553	-5.004672	0.0000006	-0.4251741	-0.1846608
month_factor9	-0.3233852	0.0611431	-5.288988	0.0000001	-0.4441073	-0.2024115
month_factor10	-0.3785233	0.0614995	-6.154906	0.0000000	-0.4996281	-0.2572052
month_factor11	-0.2205713	0.0599641	-3.678387	0.0002347	-0.3388184	-0.1019311
month_factor12	-0.3064298	0.0606558	-5.051945	0.0000004	-0.4259378	-0.1866087

both having statistically significant effects on shut in duration, suggesting that higher levels of both could have a higher influence on the decision to shut in.

Table 3: XTO Probit Model Results

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-29.4492961	11.3660902	-2.5909786	0.0095703	-51.4982248	-7.4308857
log_lag_production	0.4201253	0.0180169	23.3183501	0.0000000	0.3858429	0.4564400
log_lag_wtr	-0.3180159	0.0111933	-28.4113808	0.0000000	-0.3399123	-0.2965275
lat	-0.1558617	0.1110067	-1.4040748	0.1602966	-0.3744817	0.0615149
long	-0.3393488	0.1076207	-3.1531923	0.0016150	-0.5480061	-0.1304764
month_factor2	0.0423838	0.1675657	0.2529386	0.8003157	-0.2888975	0.3736428
month_factor3	-0.3483678	0.1563117	-2.2286741	0.0258356	-0.6550511	-0.0419241
month_factor4	-0.1357401	0.1532032	-0.8860132	0.3756104	-0.4353140	0.1646282
month_factor5	0.0199636	0.1516706	0.1316245	0.8952813	-0.2753816	0.3167287
month_factor6	-0.3568208	0.1572193	-2.2695741	0.0232334	-0.6645334	-0.0494442
month_factor7	-0.1990870	0.1541787	-1.2912746	0.1966085	-0.5017508	0.1040667
month_factor8	-0.1520582	0.1542097	-0.9860482	0.3241094	-0.4551221	0.1515342
month_factor9	-0.2219580	0.1558637	-1.4240515	0.1544316	-0.5292906	0.0854349
month_factor10	-0.1737198	0.1558263	-1.1148300	0.2649233	-0.4799634	0.1327766
month_factor11	0.0531922	0.1530058	0.3476480	0.7281046	-0.2481056	0.3556945
month_factor12	-0.0478677	0.1553500	-0.3081277	0.7579852	-0.3533631	0.2583939

Table 4: Combined Probit Model Results

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-86.4000346	3.0691437	-28.1511854	0.0000000	-92.3696392	-80.4470563
log_lag_production	0.5067034	0.0107070	47.3245268	0.0000000	0.4865749	0.5272903
log_lag_wtr	-0.3508890	0.0057958	-60.5422178	0.0000000	-0.3620115	-0.3398688
lat	0.2793195	0.0276284	10.1098767	0.0000000	0.2253366	0.3337061
long	-0.6860963	0.0250843	-27.3516167	0.0000000	-0.7348365	-0.6375201
month_factor2	0.1469587	0.0604562	2.4308308	0.0150642	0.0276932	0.2664221
month_factor3	-0.3143345	0.0571716	-5.4980875	0.0000000	-0.4266459	-0.2018380
month_factor4	-0.1904454	0.0560574	-3.3973268	0.0006805	-0.3006028	-0.0799771
month_factor5	-0.0527941	0.0553816	-0.9532792	0.3404486	-0.1615284	0.0563452
month_factor6	-0.2976808	0.0567684	-5.2437774	0.0000002	-0.4092742	-0.1858563
month_factor7	-0.2957980	0.0567755	-5.2099603	0.0000002	-0.4074578	-0.1839103
month_factor8	-0.2807084	0.0566418	-4.9558493	0.0000007	-0.3922148	-0.1689673
month_factor9	-0.3045500	0.0568669	-5.3554891	0.0000001	-0.4167529	-0.1921482
month_factor10	-0.3472183	0.0571575	-6.0747619	0.0000000	-0.4596694	-0.2345936
month_factor11	-0.1830772	0.0557840	-3.2818961	0.0010311	-0.2930068	-0.0728243
month_factor12	-0.2698837	0.0564279	-4.7828039	0.0000017	-0.3810118	-0.1585006

The effects of well geography also play a role, with noticeable effects with both firms (Tables 2-3), though latitude seems to not be significant for XTO (Table 3)

The timing of the COVID-19 pandemic seemed to more heavily affect Whiting than XTO, with every month other than may being significant in prediction, with March of 2020 being extra significant (Table 2) whereas XTO has only March of 2020 as a significant predictor out of all the months (Table 3), which lines up with the start of state issued lockdowns.

Overall, these results suggest that shut in behaviors is influenced by both the external market conditions as

well as the production for the firms. Pooling the firms together may obscure differences in behavior, as we saw with the difference in significance with months of the two firms.

Logit models were estimated as a robustness check. The results were consistent with the probit models, so the report focuses on the probit results.

Table 5: Likelihood Ratio Test for Firm Pooling

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
28561	17904.68	NA	NA	NA
28545	17825.03	16	79.646	0

To formally assess whether it is appropriate to create a single, combined model that can be applied to both firms, a likelihood ratio test was conducted comparing a pooled model to a model with interaction terms, allowing coefficients to vary by firm (Table 8).

The results show a significant improvement in model fit when allowing for firm-specific effects (Deviance = 79.646, $p < 1.927e-10$). This indicates that the relationship between well characteristics and shut-in decisions differs across firms. Therefore, it is not statistically appropriate to pool Whiting and XTO wells into a single model, and separate models provide a more accurate representation of firm behavior.

Cox Proportional Hazards Model

Table 6: Cox Proportional Hazards Model Results

term	estimate	std.error	statistic	p.value	conf.low	conf.high
log1p(production)	1.0364266	0.0239894	1.491442	0.1358455	0.9888234	1.0863215
log1p(wtr)	1.0439012	0.0241787	1.776971	0.0755729	0.9955853	1.0945618
lat	0.7861371	0.0537941	-4.473060	0.0000077	0.7074711	0.8735502
long	0.8569129	0.0521314	-2.962110	0.0030554	0.7736817	0.9490979
companyXTO ENERGY INC	0.8327069	0.0591560	-3.094759	0.0019697	0.7415467	0.9350737

To analyze the duration of shut-in periods and identify factors that influence the speed of their return, we employed Cox models (Table 9). The hypotheses for this model are presented below:

H_0 : Well characteristics have no effect on how long wells remain shut in

H_1 : Well characteristics significantly affect the duration of shut-in

The results show that neither production nor wastewater were significant predictors for recovery, though wastewater production was close to being significant with ($p = 0.076$), indicating that wells with higher wastewater may have higher chance, but is not important enough to note.

The latitudes and longitudes seem to play a bigger part with with ($p = 0.00001$) for lat and ($p = 0.003$) for Longitude respectively. This implies that the geographical location of these wells may play a part in their rate of recovery

Firm-level differences are also important. Wells operated by XTO have a significantly lower hazard rate compared to Whiting ($p < 0.01$), suggesting that XTO wells tend to remain shut in for longer periods.

Overall, these results suggest that shut-in duration is influenced by both well characteristics and firm behavior. While most shut-ins are temporary, the time required for wells to return to production varies systematically across wells and firms.

Recommendations

Question 1: Do shut-in behaviors differ between firms, particularly between Whiting and XTO in the Bakken region?

Shut-in behavior does differ between firms, with Whiting seemingly more effected overall by the initial COVID-19 shock with significance across all months. The probit model results show that the probability of shut-in increased significantly for both firms in March, indicating a strong common response to market conditions despite initial responses.

The influences of well characteristics and geographic location does seem identical with the two firms, though with latitude being not significant for XTO we can say that XTO might not been as effected by their location as much as Whiting.

Question 2: How long do wells remain shut in before returning to production, and are shut-in durations influenced by firm behavior and well characteristics?

The Cox model results suggest that shut-in duration is influenced by both well characteristics and firm behavior. Most shut-in episodes are relatively short, with many wells returning to production within about one month. However, among wells that were shut in, the geographical locations in latitude and longitude seem to be significant predictors in time before return.

The model also shows that XTO wells tended to remain shut in longer than Whiting wells, indicating that firm-level differences played a role in recovery timing. Overall, shut-ins were generally temporary, but the duration of shut-ins was not random; it varied based the firm's behaviors and the well's location.

Statistical Questions:

Question 1: Can a probit model adequately explain shut-in decisions based on observable well characteristics?

The probit model provides evidence that shut-in decisions can be partially explained by observable well characteristics and geographical data, though the way each firm reacted to shut ins throughout the months were different. In particular, the months of the year played as a significant predictor for Whiting but not for XTO for all except March, where it was significant for both.

Question 2: Is it statistically appropriate to combine Whiting and XTO wells into a single model, or should they be modeled separately?

The comparison between firm-specific models and the combined model suggests that it may not be appropriate to simply pool Whiting and XTO wells into a single model. Differences in coefficient significance and interpretation—particularly for production and geographic variables—indicate that firm behavior is heterogeneous. As a result, separate models provide a more accurate representation of shut-in decision-making across firms.

Additional Considerations

Limitations

This analysis is subject to several limitations. The definition of a shut-in is based on a production threshold (less than 50% of the previous month), which was validated through exploratory analysis. While this provides a consistent and data-driven rule, it may not perfectly capture all true operational shut-ins. Some wells may experience production declines due to mechanical issues or natural depletion rather than deliberate firm decisions. While EDA suggests a relationship between production and shut-in behavior, the regression results indicate that this effect is not statistically significant for all firms.

In addition, the Cox model does not explicitly account for censoring of wells that did not return to production within the observation window.

Finally, the dataset is limited to two firms operating in the Bakken region, which may restrict the generalizability of the results to other firms or geographic areas. The use of monthly data also limits the ability to capture shorter-term production adjustments that may occur within a given month.

Concerns and Future Research

A key concern is the absence of direct economic variables, such as oil prices, operating costs, or firm-level financial data. While the timing of shut-ins aligns with the COVID-19 shock, the analysis cannot directly quantify the effect of price changes on shut-in decisions. Future research could incorporate external data sources, such as oil price time series or firm financial indicators, to better isolate the drivers of shut-in behavior. Expanding the analysis to include additional firms or regions would also improve the robustness of the findings.

In addition, while this study includes a Cox proportional hazards model to analyze shut-in duration, further extensions—such as incorporating time-varying covariates or more detailed operational data—could provide deeper insight into the dynamics of shut-ins over time.

Technical Comments

Several modeling choices may influence the results. The use of log transformations helps reduce skewness in production variables, but may also affect interpretability. The logistic and probit models assume independence across observations, which may not fully account for repeated measurements from the same well over time.

Furthermore, while separate models were estimated for each firm, unobserved heterogeneity may still be present at the well level. Incorporating fixed effects or random effects models could better account for these differences.

Finally potential measurement error in reported production values may introduce additional uncertainty into the estimates.

References

1. U.S. Energy Information Administration. Petroleum & Other Liquids: RBRTE (Monthly Data). U.S. Department of Energy. <https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=PET&s=RBRTE&f=M>

Technical Appendix

Detailed information and a copy of code and or software results. Additional graphs and supporting figures may also be placed in the appendix.

R Script

```
# clean up & set default chunk options
rm(list = ls())
knitr::opts_chunk$set(echo = FALSE)
options(dplyr.summarise.inform = FALSE)

#packages
library(readxl)
library(dplyr)
library(janitor)
library(ggplot2)
library(kableExtra)
library(purrr)
library(mosaic)
library(stringr)
library(knitr)
library(broom)
library(survival)

#download data
read_and_clean <- function(file_path) {
  df <- read_excel(file_path, col_types = "text", na = c("", "NA")) %>%
    clean_names() %>%
    mutate(report_date = as.character(report_date),
           across(c(oil, gas, wtr, days, flared), ~as.numeric(gsub(",", "", trimws(as.character(.))))),
           api_wellno = as.character(api_wellno))

  file_name <- basename(file_path)
  date_str <- str_extract(file_name, "\\d{4}_\\d{2}")
  year <- as.integer(substr(date_str, 1, 4))
  month <- as.integer(substr(date_str, 6, 7))

  df %>%
    mutate(year = year, month = month, date = as.Date(paste(year, month, "15", sep = "-")))
}

files <- list.files("~/Desktop/STAT 470W/Oil and Gas Project/data", pattern = "\\..xlsx$", full.names = TRUE)
files <- files[!grepl("^~\\$", basename(files))]
df_all <- lapply(files, read_and_clean) %>%
  bind_rows()
#change the file path to where you keep all your data eg. "~/Downloads/"

df_horizontal_bakken <- df_all %>%
  filter(company %in% c(
    "WHITING OIL AND GAS CORPORATION",
    "XTO ENERGY INC")) %>%
  filter(grepl("H$", well_name)) #only horizontal wells

#glimpse(df_horizontal_bakken)
#colSums(is.na(df_horizontal_bakken[, c("oil", "gas", "wtr", "days", "flared"))])
variables_table <- data.frame(
  Variable = c(
```

```

    "api_wellno", "company", "month", "date",
    "oil", "gas", "wtr", "days", "flared",
    "lat", "long", "well_name",
    "production", "shut_in", "recovered"
  ),
  Type = c(
    "Identifier", "Categorical", "Time", "Date",
    "Numeric", "Numeric", "Numeric", "Numeric", "Numeric",
    "Numeric", "Numeric", "Character",
    "Numeric (derived)", "Binary (derived)", "Binary (derived)"
  ),
  Description = c(
    "Unique well identifier",
    "Operator name",
    "Month of observation",
    "Date of observation",
    "Monthly oil production (barrels)",
    "Monthly gas production (MCF)",
    "Water production (wastewater)",
    "Number of production days",
    "Gas flared",
    "Latitude of well",
    "Longitude of well",
    "Well name (used to identify horizontal wells)",
    "Total production (oil + gas/6)",
    "Indicator for shut-in (<50% of previous month)",
    "Indicator for recovery (>=75% of pre-shut production)"
  )
)

kable(
  variables_table,
  format = "latex",
  booktabs = TRUE,
  caption = "Summary of Variables Used in the Analysis",
  col.names = c("Variable Name", "Type", "Description"),
  align = c("l", "c", "l")
) %>%
  kable_styling(latex_options = c("hold_position"), full_width = FALSE)
df_horizontal_bakken <- df_horizontal_bakken %>%
  mutate(production = oil + (gas / 6)) %>%
  arrange(api_wellno, month) %>%
  group_by(api_wellno) %>%
  mutate(
    lag_production = lag(production),
    prod_ratio = ifelse(lag_production > 0, production / lag_production, NA),

    # Shut-in definition
    shut_in = ifelse(!is.na(prod_ratio) & prod_ratio < 0.5, 1, 0),

    # Recovery definition (75% of pre-shut production)
    pre_shut_prod = lag_production,
    recovered = ifelse(production >= 0.75 * pre_shut_prod, 1, 0)
  )

```

```

) %>%
  ungroup()
# Monthly production by company
monthly_prod <- df_horizontal_bakken %>%
  group_by(date, company) %>%
  summarise(total_production = sum(production, na.rm = TRUE), .groups = "drop") %>%
  arrange(company, date)

ggplot(monthly_prod, aes(x = date, y = total_production, color = company, group = company)) +
  geom_line() +
  labs(title = "Monthly Production by Firm", x = "Month", y = "Total Production") +
  theme(axis.text.x = element_text(angle = 90, hjust = 0.5)) +
  scale_x_date(date_breaks = "1 month") +
  geom_vline(xintercept = as.Date("2020-03-15"), color = "red", linetype = "dashed")
# Shut-in rate
monthly_shutin <- df_horizontal_bakken %>%
  group_by(date, company) %>%
  summarise(shut_in_rate = mean(shut_in, na.rm = TRUE), .groups = "drop")

ggplot(monthly_shutin, aes(x = date, y = shut_in_rate, color = company, group = company)) +
  geom_line() +
  labs(title = "Shut-in Rate Over Time", x = "Month", y = "Shut-in Rate") +
  theme(axis.text.x = element_text(angle = 90, hjust = 0.5)) +
  scale_x_date(date_breaks = "1 month") +
  geom_vline(xintercept = as.Date("2020-03-15"), color = "red", linetype = "dashed")
# Identify shut-in episodes
df_horizontal_bakken <- df_horizontal_bakken %>%
  arrange(api_wellno, date) %>%
  group_by(api_wellno) %>%
  mutate(shut_group = cumsum(shut_in == 1 & lag(shut_in, default = 0) == 0)) %>%
  ungroup()

# Duration per episode
duration_data <- df_horizontal_bakken %>%
  filter(shut_in == 1) %>%
  group_by(api_wellno, shut_group) %>%
  summarise(duration = n())

# Histogram of durations
ggplot(duration_data, aes(x = duration)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Shut-in Durations") +
  scale_x_continuous(breaks = seq(0, 20, by = 1)) +
  scale_y_continuous(breaks = seq(0, 2000, by = 200))

#duration_data %>%
# dplyr::group_by(duration) %>%
# dplyr::summarize(count = dplyr::n())
# Subset to post-COVID period
df_analysis <- df_horizontal_bakken %>%
  filter(date >= as.Date("2020-03-01"))

# Create production quartiles

```

```

df_analysis <- df_analysis %>%
  mutate(prod_group = ntile(production, 4))

# Shut-in rate by production quartile
prod_analysis <- df_analysis %>%
  group_by(prod_group, company) %>%
  summarise(
    shut_in_rate = mean(shut_in, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(prod_analysis, aes(x = factor(prod_group), y = shut_in_rate, fill = company)) +
  geom_bar(stat = "identity", position = position_dodge(width = 0.9)) +
  labs(
    title = "Shut-in Rate by Production Quartile (Post-Feb 2020)",
    x = "Production Quartile (1 = lowest, 4 = highest)",
    y = "Shut-in Rate"
  )

#df_analysis %>%
#  group_by(shut_in) %>%
#  summarise(
#    mean_prod = mean(production, na.rm = TRUE),
#    median_prod = median(production, na.rm = TRUE),
#    n = n(),
#    .groups = "drop"
#  )
# Boxplot: Production vs Shut-in status
ggplot(df_analysis, aes(x = factor(shut_in), y = production, fill = company)) +
  geom_boxplot() +
  scale_y_log10() +
  labs(
    title = "Production Distribution by Shut-in Status (Log Scale)",
    x = "Shut-in (0 = No, 1 = Yes)",
    y = "Production (log scale)"
  )

model_data <- df_horizontal_bakken %>%
  group_by(api_wellno) %>%
  mutate(
    lag_wtr = lag(wtr),
    shut_in_before_march = any(month < 3 & year == 2020 & shut_in == 1, na.rm = TRUE),
    log_lag_production = log1p(lag_production),
    log_lag_wtr = log1p(lag_wtr)
  ) %>%
  ungroup() %>%
  filter(
    date >= as.Date("2020-03-15"),
    shut_in_before_march == FALSE,
    !is.na(shut_in),
    !is.na(log_lag_production),
    !is.na(log_lag_wtr),
    !is.na(lat),
    !is.na(long)
  )

```

```

) %>%
mutate(
  company = factor(company),
  month_factor = factor(month),
  lat = as.numeric(lat),
  long = as.numeric(long)
)

# Probit models

# Whiting only
probit_whiting <- glm(
  shut_in ~ log_lag_production + log_lag_wtr + lat + long + month_factor,
  data = model_data %>% filter(company == "WHITING OIL AND GAS CORPORATION"),
  family = binomial(link = "probit")
)

# XTO only
probit_xto <- glm(
  shut_in ~ log_lag_production + log_lag_wtr + lat + long + month_factor,
  data = model_data %>% filter(company == "XTO ENERGY INC"),
  family = binomial(link = "probit")
)

# Combined Whiting and XTO
probit_combined <- glm(
  shut_in ~ log_lag_production + log_lag_wtr + lat + long + month_factor,
  data = model_data,
  family = binomial(link = "probit")
)

tidy(probit_whiting, conf.int = TRUE) %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Whiting Probit Model Results"
  ) %>%
  kable_styling(latex_options = c("hold_position"), full_width = FALSE)

tidy(probit_xto, conf.int = TRUE) %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "XTO Probit Model Results"
  ) %>%
  kable_styling(latex_options = c("hold_position"), full_width = FALSE)

tidy(probit_combined, conf.int = TRUE) %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Combined Probit Model Results"
  ) %>%

```

```

kable_styling(latex_options = c("hold_position"), full_width = FALSE)
# whiting only:
logit_whiting <- glm(shut_in ~ log_lag_production + log_lag_wtr + lat + long + month_factor,
  data = model_data %>% filter(company == "WHITING OIL AND GAS CORPORATION"),
  family = binomial(link = "logit"))
# xto only:
logit_xto <- glm(shut_in ~ log_lag_production + log_lag_wtr + lat + long + month_factor,
  data = model_data %>% filter(company == "XTO ENERGY INC"),
  family = binomial(link = "logit"))
# whiting and xto:
logit_combined <- glm(shut_in ~ log_lag_production + log_lag_wtr + lat + long + month_factor,
  data = model_data,
  family = binomial(link = "logit"))
model_pooled <- glm(
  shut_in ~ log_lag_production + log_lag_wtr + lat + long + month_factor,
  data = model_data,
  family = binomial(link = "probit")
)
model_interaction <- glm(
  shut_in ~ (log_lag_production + log_lag_wtr + lat + long + month_factor) * company,
  data = model_data,
  family = binomial(link = "probit")
)
lrt_result <- anova(model_pooled, model_interaction, test = "Chisq")

kable(
  lrt_result,
  format = "latex",
  booktabs = TRUE,
  caption = "Likelihood Ratio Test for Firm Pooling"
) %>%
  kable_styling(latex_options = c("hold_position"), full_width = FALSE)
# Fix lat/long type before survival model
df_horizontal_bakken <- df_horizontal_bakken %>%
  mutate(
    lat = as.numeric(lat),
    long = as.numeric(long),
    wtr = as.numeric(wtr)
  )
# create survival data
surv_data <- df_horizontal_bakken %>%
  arrange(api_wellno, date) %>%
  group_by(api_wellno) %>%
  mutate(
    shut_group = cumsum(shut_in == 1 & lag(shut_in, default = 0) == 0)
  ) %>%
  ungroup() %>%
  filter(shut_in == 1) %>%
  group_by(api_wellno, shut_group) %>%
  summarise(
    duration = n(),
    company = first(company),

```



```

    production = first(production),
    wtr = first(wtr),
    lat = first(lat),
    long = first(long),
    .groups = "drop"
  )

# Cox model
cox_model <- coxph(
  Surv(duration) ~ log1p(production) + log1p(wtr) + lat + long + company,
  data = surv_data
)

tidy(cox_model, exponentiate = TRUE, conf.int = TRUE) %>%
  kable(
    format = "latex",
    booktabs = TRUE,
    caption = "Cox Proportional Hazards Model Results"
  ) %>%
  kable_styling(latex_options = c("hold_position"), full_width = FALSE)

```